

Circuit Analysis (Lab # 6)

Experiment 6: VERIFICATION OF KIRCHHOFFS LAWS

Name:

Registration No:

Date:

Grade and Signature:

CLO2	Use with identification the circuit components, breadboards, multimeters, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
CLO3				
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)			Total marks (10)	

Objective

To verify Kirchhoff's current law and Kirchhoff's voltage law for the given Circuit.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	330ohm, 220ohm 1kohm	3

Description

KCL: The algebraic sum of the currents meeting at a node is equal to zero.

KVL: In any closed path / mesh, the algebraic sum of all the voltages is zero.

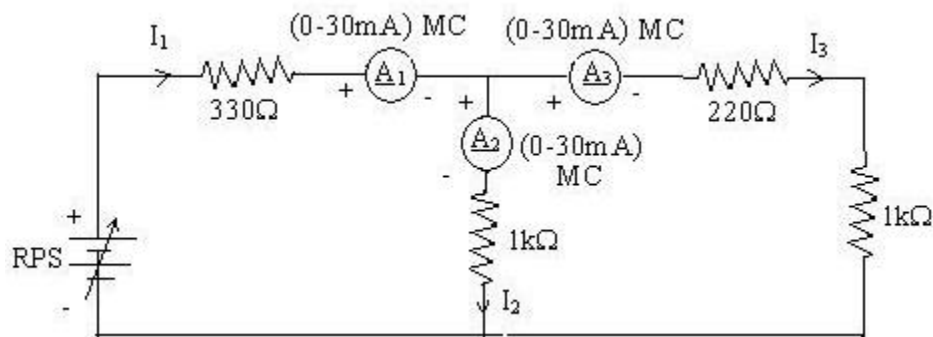
Precautions:

1. Voltage control knob should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position.

Procedure for KCL:

1. Give the connections as per the circuit diagram.
2. Set a value in RPS.
3. Note down the corresponding readings using multimeter
4. Repeat the same for different voltages

Circuit - KCL



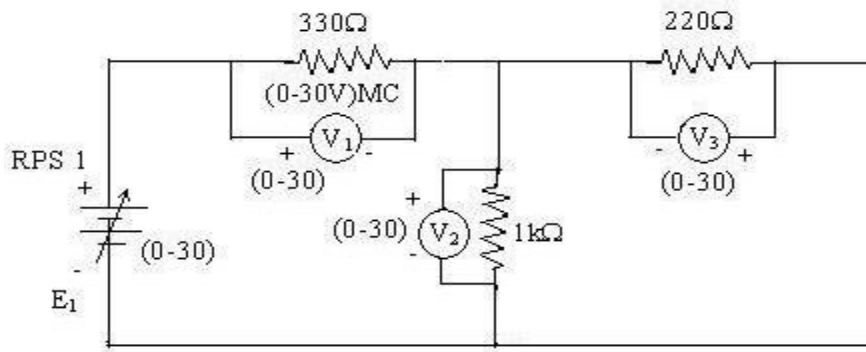
KCL - Practical Values:

Sl. No.	Voltage E	Current			$I_1 = I_2 + I_3$
		I_1	I_2	I_3	
	Volts	mA	mA	mA	mA
1	5				
2	15				
3	25				

Procedure for KVL:

1. Give the connections as per the circuit diagram.
2. Set a value in RPS.
3. Note all the voltage reading
4. Repeat the same for different voltages

Circuit - KVL



KVL - Practical Values

Sl. No.	RPS	Voltage			$E1 = V1 + V2$
	E1	V1	V2	V3	
	V	V	V	V	V
1	5				
2	10				
3	15				

Result:

Thus, Kirchhoff's voltage law and Kirchhoff's current law are verified practically by performing this experiment.

Post Lab Questions

1. Define Kirchhoff's current law.

2. Define Kirchhoff's voltage law.

3. What are the applications of regulated power supply?

4. What are the limitations of KCL?
